

# CS 4782 Homework 4

## Problem 1: Generative Models (24 points)

With the price of on-campus parking skyrocketing, you devised a scheme to print fake parking permits. To create these fake parking permits, you plan to apply what you learned about in your deep learning class this week: generative models.

Cornell has 30 types of permits, each allowing access to different parking lots. You decide to try out three different generative vision models to see which one will give you the best results in generating these various permits. However, you notice that each of the models you try out have drawbacks. Match each of the challenges listed below with the generative model they most likely correspond to and explain your answer in 2-3 sentences.

**1(a).** Although there are 30 different types of parking permits, you noticed that **model #1** only generates 3 of these permit types. (8 points)

**Solution: Model #1** is mostly likely a Generative Adversarial Network (GAN). A common issue with training GANs is mode collapse, because during training the generator finds one (or several) that fool the discriminator and therefore fixates on these few examples, therefore failing to capture the full diversity represented in the full data distribution.

**1(b).** The next model you try, **model #2**, seems to always generate blurry-looking parking permits. (8 points)

**Solution: Model #2** is mostly likely a Variational Autoencoder (VAE). A common issue seen in VAEs is blurry outputs. This is due to the training objective of VAEs incentivizing the model to output very likely samples. However, this leads to VAEs often outputting an "averaged" image, which appears blurry, as individual, sharp details are unique to a single sample and therefore very unlikely.

This is similar to the graphics showing what the average person from each country looks like, usually these faces are very blurry/fuzzy.

**1(c).** You finally find a model, **model #3**, that generates high quality, different-looking parking permits. You want to use **model #3** to mass-produce fake parking permits and sell them at a discounted price. However, you quickly realize that **model #3** is extremely slow at generating samples, which hinders your scheme to become rich. (8 points)

**Solution: Model #3** is most likely a Diffusion Model. Although diffusion models achieve high sample quality and diversity, they unfortunately have a very slow sampling speed due to the many denoising steps and working in pixel-space (at least in the original diffusion models).

## Problem 2: ELBO (40 points)

In this question, we will work on an alternative derivation for the **Evidence Lower Bound (ELBO)**, where the "evidence" is the log likelihood of the observed data. The ELBO is a key component in variational

inference and variational autoencoders (VAEs), as it provides a tractable way to approximate the intractable log marginal likelihood.

Your task is to derive the ELBO expression in terms of observed  $x$ , latent  $z$ , and  $q_\phi(z|x)$ , which is the variational distribution that seeks to optimize the true posterior  $p(z|x)$  by optimizing  $\phi$ . You have already seen an alternative derivation of the ELBO in class, but this approach follows a different path.

1. First, we need to express the log marginal likelihood in terms of the joint distribution. [Hint: This involves marginalizing out the latent variable using integration.] (2 points)

$$\log p(x) = \log \int p(x, z) dz$$

2. Next, we introduce the variational distribution  $q_\phi(z|x)$  in a way that does not change the expression. [Hint: Think about multiplying by an identity.] (2 points)

$$\log p(x) = \log \int \frac{p(x, z) q_\phi(z|x)}{q_\phi(z|x)} dz$$

3. Now, use the definition of expectation to simplify the integral further. (3 points)

$$\log p(x) = \log \mathbb{E}_{q_\phi(z|x)} \left[ \frac{p(x, z)}{q_\phi(z|x)} \right]$$

4. Why is the expectation with respect to the variational distribution a useful quantity? Explain in 2-3 sentences. (5 points)

The expectation with respect to the variational distribution,  $\mathbb{E}_{q_\phi(z|x)} \left[ \frac{p(x, z)}{q_\phi(z|x)} \right]$ , is a useful quantity because it provides a tractable way to approximate the intractable log marginal likelihood,  $\log p(x)$ . The variational distribution  $q_\phi(z|x)$  is designed to be a close approximation to the true posterior  $p(z|x)$ , but it is possible to sample from and optimize. By expressing the log marginal likelihood as an expectation under the variational distribution, we can develop optimization techniques to find the best variational parameters  $\phi$  that maximize this quantity, which in turn provides a tight lower bound on the log marginal likelihood.

5. Finally, we can convert this expression into a lower bound on the log marginal likelihood using a specific inequality. [Hint: Consider Jensen's inequality.] (3 points)

$$\log p(x) \geq \mathbb{E}_{q_\phi(z|x)} \left[ \log \frac{p(x, z)}{q_\phi(z|x)} \right]$$

6. What is the significance of having a lower bound on the log marginal likelihood? Explain in 2-3 sentences. (5 points)

Having a lower bound on the log marginal likelihood,  $\log p(x)$ , is significant because the log marginal likelihood itself is often intractable for complex probabilistic models. By deriving a lower bound, such as the ELBO, we can optimize this bound instead of the intractable quantity. Maximizing the ELBO with respect to the variational parameters  $\phi$  and the model parameters  $\theta$  will simultaneously maximize the lower bound and provide a tight approximation to the log marginal likelihood. This allows us to perform approximate inference and learning in complex probabilistic models, such as variational autoencoders (VAEs), where the true posterior and marginal likelihood are intractable.

5. Now, we can extend the bound you found in step 4 for VAEs. Recall the factorization assumptions made in VAEs (e.g., the conditional independence of  $x$  and  $z$  given the latent variables). Show that the final inequality term is equal to  $\mathbb{E}_{q_\phi(z|x)} [\log p_\theta(x|z)] - D_{KL}(q_\phi(z|x) || p(z))$  with three steps: using the chain rule of probability, splitting the expectation, and using the definition of KL divergence. (10 points)

$$\begin{aligned}
\log p(x) &\geq \mathbb{E}_{q_\phi(z|x)} \left[ \log \frac{p(x, z)}{q_\phi(z|x)} \right] \\
&= \mathbb{E}_{q_\phi(z|x)} \left[ \log \frac{p_\theta(x|z)p(z)}{q_\phi(z|x)} \right] \\
&= \mathbb{E}_{q_\phi(z|x)} [\log p_\theta(x|z)] + \mathbb{E}_{q_\phi(z|x)} \left[ \log \frac{p(z)}{q_\phi(z|x)} \right] \\
&= \mathbb{E}_{q_\phi(z|x)} [\log p_\theta(x|z)] - D_{KL}(q_\phi(z|x) \| p(z))
\end{aligned}$$

5.1 Interpret the two terms in the final ELBO expression in the context of VAEs. How does this relate to the concepts of reconstruction error and regularization? Explain in 2-3 sentences. (10 points)

In the final ELBO expression,  $\mathbb{E} q_\phi(z|x) [\log p_\theta(x|z)] - D_{KL}(q_\phi(z|x) \| p(z))$ , the first term represents the expected reconstruction error, and the second term represents the regularization term based on the Kullback-Leibler (KL) divergence between the variational distribution and the prior distribution.

In the context of VAEs, the first term encourages the model to reconstruct the input data  $x$  well from the latent representations  $z$  sampled from the variational distribution  $q_\phi(z|x)$ . Maximizing this term improves the reconstruction quality of the VAE.

The second term, the KL divergence between  $q_\phi(z|x)$  and  $p(z)$ , acts as a regularizer that encourages the variational distribution to be close to the prior distribution  $p(z)$ . This regularization term prevents the VAE from overfitting to the training data and helps in learning a smooth and continuous latent representation space.

By optimizing both terms simultaneously, the VAE balances the trade-off between reconstruction accuracy and regularization, leading to good generalization performance and the ability to generate new samples from the learned latent space.

### Problem 3: Diffusion Derivations (36 points)

1. To derive the loss function, as done in class, we want to measure the distance between the probability distribution

$$q(\mathbf{z}_{t-1} | \mathbf{z}_t, \mathbf{x}), p_\theta(\mathbf{z}_{t-1} | \mathbf{z}_t)$$

for each  $t$ , and learn models  $\mu_\theta, \Sigma_\theta$  such that the distance between the above two probability distributions is small. In fact, this is exactly how we learn the reverse process, which enables us to sample using diffusion models.

To this end, we measured the distance between these probability distributions using the *KL-divergence*  $D_{KL}(P \| Q)$ . This represents some form of distance between the probability distributions  $P$  and  $Q$ . In class, recall that we cited a formula between  $P$  and  $Q$  when  $P$  and  $Q$  were multivariate Gaussians. Eventually, this gave us the loss

$$L(\theta) = \mathbb{E}_{\mathbf{x}, t} [\|\tilde{\mu} - \mu_\theta\|^2]$$

where we have  $p_\theta(\mathbf{z}_{t-1} | \mathbf{z}_t) \sim \mathcal{N}(\mu_\theta(\mathbf{z}_t, t), \Sigma_\theta(\mathbf{z}_t, t))$ ,  $q(\mathbf{z}_{t-1} | \mathbf{z}_t, \mathbf{x}) \sim \mathcal{N}(\tilde{\mu}(\mathbf{z}_t, \mathbf{x}), \tilde{\Sigma}(\mathbf{z}_t, \mathbf{x}))$ , and we try to learn  $p_\theta(\mathbf{z}_{t-1} | \mathbf{z}_t) \approx q(\mathbf{z}_{t-1} | \mathbf{z}_t, \mathbf{x})$ . However, how is the KL divergence defined and why is it a distance? Here we answer this.

This is defined as

$$D_{KL}(P \| Q) = \mathbb{E}_P \left( \log \frac{P}{Q} \right)$$

For discrete distributions, this will be a sum, and for continuous distributions, this will be an integral.

- (a) The KL divergence is supposed to represent some form of distance between the probability distributions  $P, Q$ . For this to make sense, it must be non-negative. Prove that  $D_{KL}(P||Q) \geq 0$ . *Hint: you may assume that  $P, Q$  have given densities  $P(x), Q(x)$ . Also, use Jensen's Inequality: for a concave function  $f$ , a probability distribution  $\pi$ , and a random variable  $X$ , we have*

$$\mathbb{E}_\pi(f(X)) \leq f(\mathbb{E}_\pi(X))$$

where the expectation is taken with respect to the probability distribution  $\pi$ . (6 points)

*Solution: log is concave, so by Jensen*

$$-D_{KL}(P||Q) = \mathbb{E}_P \log \frac{Q}{P} \leq \log \int P(x) \frac{Q(x)}{P(x)} dx = \log 1 = 0$$

We don't actually need to use the fact there are given densities, we can just use the properties of Radon-Nikodym derivative, but this is beyond the scope of this assignment.

- (b) Recall we used a formula for the KL divergence between two Gaussians in class, to create the loss function to train diffusion models. This formula was: for two multivariate Gaussian distributions in  $\mathbb{R}^d$ ,  $P \sim \mathcal{N}(\mu_1, \Sigma_1)$ ,  $Q \sim \mathcal{N}(\mu_2, \Sigma_2)$ ,

$$D_{KL}(P||Q) = \frac{1}{2} \left( \text{tr}(\Sigma_2^{-1}\Sigma_1) - d + (\mu_2 - \mu_1)^T \Sigma_2^{-1}(\mu_2 - \mu_1) + \ln\left(\frac{\det \Sigma_2}{\det \Sigma_1}\right) \right)$$

where  $\text{tr}$  denotes trace of a matrix (assuming  $\Sigma_1, \Sigma_2$  are invertible).

In the next problem, we'll derive it for the case of univariate Gaussian distributions. For this problem, consider two univariate Gaussian distributions,  $P \sim \mathcal{N}(\mu_1, \sigma_1^2)$ ,  $Q \sim \mathcal{N}(\mu_2, \sigma_2^2)$ . We have that

$$D_{KL}(P||Q) = \frac{1}{2} \left( \sigma_2^{-2}\sigma_1^2 + (\mu_2 - \mu_1)^2\sigma_2^{-2} - 1 + \ln\left(\frac{\sigma_2^2}{\sigma_1^2}\right) \right)$$

Prove that this formula for KL divergence between univariate Gaussians follows as a special case from the general formula for KL divergence between multivariate Gaussians written above. (12 points)

*Solution: Single variable Gaussians are multivariate Gaussians with  $d = 1$ , and the trace and determinant of a  $1 \times 1$  matrix is just the real number that is the matrix element. So using the formula above, we get*

$$\text{tr}(\Sigma_2^{-1}\Sigma_1) = \sigma_2^{-2}\sigma_1^2, \det \Sigma_i = \sigma_i^2$$

Also, since all these quantities are reals they commute, so

$$(\mu_2 - \mu_1)^T \Sigma_2^{-1}(\mu_2 - \mu_1) = (\mu_2 - \mu_1)^2 \sigma_2^{-2}$$

$d = 1$  so all the terms in the general formula match those of the special case.

- (c) Prove the above formula for the case of single variable Gaussians *without using the multivariate formula*: show directly that, for two univariate Gaussian distributions,  $P := \mathcal{N}(\mu_1, \sigma_1^2)$ ,  $Q := \mathcal{N}(\mu_2, \sigma_2^2)$ , we have that

$$D_{KL}(P||Q) = \frac{1}{2} \left( \sigma_2^{-2}\sigma_1^2 + (\mu_2 - \mu_1)^2\sigma_2^{-2} - 1 + \ln\left(\frac{\sigma_2^2}{\sigma_1^2}\right) \right).$$

Show all intermediate steps. (15 points)

*Solution: From the definition we can write*

$$D_{KL}(P||Q) = \int f_P(x) \log \left( \frac{\sigma_1^{-1} e^{-\frac{1}{2}(x-\mu_1)^2 \sigma_1^{-2}}}{\sigma_2^{-1} e^{-\frac{1}{2}(x-\mu_2)^2 \sigma_2^{-2}}} \right) dx$$

where  $f_P$  is the PDF of  $P$ . We can simplify the log term as

$$\frac{1}{2} \ln(\sigma_2^2/\sigma_1^2) + \frac{1}{2}((x - \mu_2)^2 \sigma_2^{-2} - (x - \mu_1)^2 \sigma_1^{-2})$$

$$= \frac{1}{2} (\ln(\sigma_2^2/\sigma_1^2) + \mu_2^2\sigma_2^{-2} - \mu_1^2\sigma_1^{-2}) - x(\mu_2\sigma_2^{-2} - \mu_1\sigma_1^{-2}) + \frac{1}{2}x^2(\sigma_2^{-2} - \sigma_1^{-2}).$$

The rest of this solution uses the facts

$$\int af_P(x)dx = a,$$

$$\int axf_P(x)dx = a\mu_1,$$

and

$$\int ax^2f_P(x)dx = a\mathbb{E}_P[X^2] = a(\text{Var}(X) + \mathbb{E}_P[X]^2) = a(\sigma_1^2 + \mu_1^2)$$

where  $X \sim P$ . Using these facts, we get

$$\begin{aligned} & \int f_P(x) \log \left( \frac{\sigma_1^{-1} e^{-\frac{1}{2}(x-\mu_1)^2\sigma_1^{-2}}}{\sigma_2^{-1} e^{-\frac{1}{2}(x-\mu_2)^2\sigma_2^{-2}}} \right) dx \\ &= \int f_P(x) \left( \frac{1}{2} (\ln(\sigma_2^2/\sigma_1^2) + \mu_2^2\sigma_2^{-2} - \mu_1^2\sigma_1^{-2}) - x(\mu_2\sigma_2^{-2} - \mu_1\sigma_1^{-2}) + \frac{1}{2}x^2(\sigma_2^{-2} - \sigma_1^{-2}) \right) dx \\ &= \int f_P(x) \left( \frac{1}{2} (\ln(\sigma_2^2/\sigma_1^2) + \mu_2^2\sigma_2^{-2} - \mu_1^2\sigma_1^{-2}) \right) dx + \int f_P(x) (x(\mu_2\sigma_2^{-2} - \mu_1\sigma_1^{-2})) dx \\ &+ \int f_P(x) \left( \frac{1}{2}x^2(\sigma_2^{-2} - \sigma_1^{-2}) \right) dx \\ &= \frac{1}{2} (\ln(\sigma_2^2/\sigma_1^2) + \mu_2^2\sigma_2^{-2} - \mu_1^2\sigma_1^{-2}) - \mu_1 (\mu_2\sigma_2^{-2} - \mu_1\sigma_1^{-2}) + \frac{1}{2}(\sigma_1^2 + \mu_1^2) (\sigma_2^{-2} - \sigma_1^{-2}) \\ &= \frac{1}{2} (\ln(\sigma_2^2/\sigma_1^2) + \mu_2^2\sigma_2^{-2} - \mu_1^2\sigma_1^{-2} - 2\mu_1\mu_2\sigma_2^{-2} + 2\mu_1^2\sigma_1^{-2} + \sigma_1^2\sigma_2^{-2} - \sigma_1^2\sigma_1^{-2} + \mu_1^2\sigma_2^{-2} - \mu_1^2\sigma_1^{-2}) \\ &= \frac{1}{2} (\ln(\sigma_2^2/\sigma_1^2) + \mu_2^2\sigma_2^{-2} - 2\mu_1\mu_2\sigma_2^{-2} + \mu_1^2\sigma_2^{-2} + \sigma_1^2\sigma_2^{-2} - 1) \\ &= \frac{1}{2} (\ln(\sigma_2^2/\sigma_1^2) + \sigma_2^{-2}(\mu_2 - \mu_1)^2 + \sigma_2^{-2}\sigma_1^2 - 1) \end{aligned}$$

as desired.